



Microplastic contamination in fish: Critical review and assessment of data quality

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ABSTRACT

Although the ingestion of microplastics by fish has been investigated extensively over the last 50 years, the quality of these ingestion studies is questionable due to the lack of standard methods for sampling, extraction, and identification of microplastics. This review aims to provide reliable results of microplastic occurrence in fish sourced from 104 published articles through a rigorous assessment of the data quality. A scoring system was performed to assess ten quality assurance/quality control criteria including sampling methods, sample size, sample processing and storage, laboratory preparation, clean air conditions, negative controls, positive controls, target component, sample treatment and polymer identification. We found that 59 out of 104 articles had scores ≤ 10 , and only two studies are identified with scores higher than 15, demonstrating the urgent need to improve the quality and reliability of fish digestion studies. Particularly, five criteria, namely, the positive controls, sampling method, clean air conditions, sample size, and polymer identification with an average score of less than 1 need attention and improvement in future studies. This review also revealed that the 104 studies considered here originated mainly from five countries, i.e., China (23%), Iran (8%), Thailand (5%), Turkey (4%), and Indonesia (4%). Among 104 studies, the major polymer type found was polyethylene, with a size range of 0–1 mm. The most frequently reported colour and shape of microplastics found in the fish were blue and fibre, respectively.

1. Introduction

Plastic residues are ubiquitous environmental contaminants and accumulate in marine, freshwater, and terrestrial ecosystems (Andrady, 2011; Horton et al., 2017). Particular attention has been given to microplastic particles (MPs, < 5 mm) (Bergmann et al., 2015) that are detected at a wide range of concentrations in marine biota (Hermesen et al., 2018), salt, drinking water and air (Zhang, Cui, et al., 2020). Microplastics are classified into two categories based on their origins, namely, primary, and secondary microplastics. Primary microplastics are defined as plastic particles that enter the environment directly, such as the cosmetics of microfibers shed from clothing and textile, while the secondary MPs originate from the degradation and breakdown of larger plastic items (Lambert & Wagner, 2018, pp. 1–23). Primary sources could contribute 15%–31% to all plastics deposited in the environment (Boucher & Friot, 2017).

Over time, MPs enter aquatic ecosystems including rivers, lakes, and ocean, providing direct exposure to fish intended for human

consumption (Smith et al., 2018). More than 690 marine species have been found to ingest MPs (Zhang, Cui, et al., 2020), which may induce physical and chemical toxicity, such as intestinal inflammation, oxidative stress, disorders of metabolome, altering the behaviour, energy reserve, and nutritional composition in fish (Qiao et al., 2019; Yin et al., 2018). Furthermore, MPs ingested by fish increases human health risk through fish consumption. Humans may experience oxidative stress, cytotoxicity, neurotoxicity, immune system disruption as revealed in animal research and *in vitro* tests (Bhuyan, 2022). Particularly, MPs have been found in placentas, meconium, infant faeces, breastmilk and infant formula (Liu et al., 2023), implying a pressing human health issue.

Considering the potential threat of MPs to animal and human health, the global status of MPs in fish has gained growing concern (Guilhermino et al., 2021). However, published studies present a considerable variability in results on the occurrence of MPs ingestion by fish, indicating a lack of comparability of results, which might be due to a lack of standardisation of methods. Hence, Hermesen et al. (2018) proposed a quality assessment method to evaluate MP ingestion studies in aquatic

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biota (e.g., mussels, fish, and zooplankton). [Hermesen et al. \(2018\)](#) summarized that the reliability of reviewed studies needed improvement to be applied confidently to risk assessment or evaluating MPs of biota and showed a requirement for better quality assurance in microplastic ingestion studies. The criteria proposed by [Hermesen et al. \(2018\)](#) have been applied to assess the data quality of studies on MPs ingested by organisms at all levels of biological organisation including fish ([Gouin, 2020](#)), as well as the quality of studies in freshwaters and drinking water ([Koelmans et al., 2019](#)). Based on the above three reviews ([Gouin, 2020](#); [Hermesen et al., 2018](#); [Koelmans et al., 2019](#)), the common criteria requiring the most improvement are positive control, laboratory preparation, clean air condition, and sample treatment. Nevertheless, the quality assessment focusing solely on the MPs ingested by fish has not been reported so far. An update on the quality assessment of data on MPs ingested by fish is essential to provide reliable information for the risk assessment of the current status of research on MPs detection in fish. To fill the current research gap, this study aims to provide information on the reliability of studies regarding MPs in fish by using quality criteria and the assessment scoring system developed by [Hermesen et al. \(2018\)](#).

Our paper is structured as follows. Firstly, we present and discuss the assessment results of 104 studies in ten criteria containing sampling methods, sample size, sample processing and storage, laboratory preparation, clean air conditions, negative controls, positive controls, target component, sample treatment and polymer identification. Thereafter, we provide an overview of the characteristics of MPs in fish, considering global distribution, polymer types, size, shapes, and colour. Finally, a general discussion is raised focusing on the limitation of the assessment system and the affecting factors of the characteristics, consequently, some recommendations. Nevertheless, the quality assessment focusing solely on the MPs. The current review will provide reliable information for the risk assessment of the current status of research on MP detection in fish and stimulate the standardisation of the MP detection.

2. Methods

The literature review was conducted by accessing the Scopus databases for studies of microplastic ingestion in fish from 2020 up until those published in March 2022, with the keywords: (microplast* OR micro-plast* OR "plastic particle*" OR microbead*) AND (fish OR seafood OR exposure OR ingestion OR intake). Initial screening was performed to limit the searched studies to be original research and published in English language. Further screening was adopted based on the following exclusion criteria through the title, abstract and/or full paper reading: (1) studies that just reported the MPs in the environment (e.g., the surface water and sediment); (2) studies that just reported MPs in other aquatic biota such as mussels and prawns instead of fish; (3) studies that just evaluate the effect of MPs in fish; (4) studies that were irrelevant to MPs were not considered. After screening, a total of 104 studies that reported original concentration data of microplastics in fish were identified.

All 104 studies are evaluated based on 10 quality assurance/quality control (QA/QC) criteria: sampling methods, sample size, sample processing and storage, laboratory preparation, clean air conditions, negative controls, positive controls, target component, sample treatment and polymer identification ([Hermesen et al., 2018](#)). A summary of the 10 QA/QC criteria is presented in [Table S1](#). Each criterion is assigned a score of 2 (reliable without restriction), 1 (somewhat reliable, with restrictions) or 0 (unreliable). The scores per individual study are provided in [Table S2](#). Generally, if a study shows positive value in any criteria, it is considered sufficiently reliable.

For each study the following characteristics were recorded in tabular form ([Table S3](#)): Location, Digestion agent, Pore size of filter paper (μm), Detection methods, Major polymer types, Major size (mm), Major shape, Major colour, Major size range (mm), Detection rate (percentage of fish containing MPs), Sources (entire fish, gastrointestinal tract, muscle,

liver, gill, viscera, gonads, or brain). Major polymer type, shape and colour represent the dominant type, shape and colour of MPs found in fish.

3. Results and discussion

3.1. Publications by journals

[Fig. 1](#) summarises the selected studies investigating MPs in fish in terms of by journal. Among all the journals, *Marine Pollution Bulletin* topped output (30 publications), followed by *Science of The Total Environment* (14), *Environmental Pollution* (13), *Chemosphere* (7) and *Environmental Science and Pollution Research* (6). The impact factors (2023) of these 5 journals are higher than 5. The journals span different disciplines including Environmental Sciences, Marine & Freshwater Biology, Toxicology, Engineering, Multidisciplinary Sciences, Water Resources, Occupational Health, Oceanography, and Zoology. The multidisciplinary journals indicated the engagement of researchers from various scholarly disciplines, implying that multidisciplinary approaches are required to address this problem.

3.2. Quality assessment

While the scoring system was not necessarily developed to provide an absolute judgment about an individual paper, it acts as an indicator of the reliability of the data reported in a study. Scoring based on this quality assessment implies in general that substantial improvements are needed in investigating MP ingestion in fish. [Fig. 2](#) displays the average scores of 10 criteria that are further clustered into the four main categories, i.e., sampling, contamination mitigation, sample purification/handling, and chemical analysis. The first category (criteria 1–3) evaluates the sampling collection and storage, which can affect the results of microplastic detection in fish samples. The average value of accumulated scores for this subset across all studies is 2.9 out of a total possible 6. The accumulated scores for each category represent a sum of scores of the corresponding criteria. Thirty-eight studies showed positive scores for each quality criterion. The second subset of the assessment criteria (criteria 4–6) relates to the control of the laboratory environment which can introduce MP contamination to the samples. For this class, the average value of accumulated scores is 2.8. Fifty-four studies have positive values for each criterion. The third subset is associated with the effectiveness of the sampling treatment and detection method, with an average accumulated score of 3.2. Only eight studies present positive scores for all criteria in this category. The fourth subset discusses polymer identification (criteria 10) with an average score of 1.0 and 77 papers reporting positive values. Compared with the other three categories, in the third subset, the lowest number of studies show positive scores for each criterion, indicating the main quality deficiencies in MP/fish studies relate to chemical analysis.

In the following paragraphs, the quality assessment of 104 studies is further detailed and discussed based on each criterion.

3.2.1. Sampling method

The sampling method considers the influence of the sampling date, location, depth, gear and mesh size on the MP concentrations in fish, as these factors can affect the results of microplastic detection in biota samples ([Hermesen et al., 2018](#)). For instance, sampling date is important as microplastic pollution is affected by certain seasonal factors ([Wang et al., 2020](#)). The wind, surface current and rainfall play an important role in carrying marine debris, especially during the monsoon season ([James et al., 2020](#)). Consequently the number of fish detected with MP was significantly higher in the wet season ([Merga et al., 2020](#)) and monsoon season ([James et al., 2020](#)). Sampling depth is also important as fish feeding on sediment might be exposed to more plastic particles than surface feeding ones ([Merga et al., 2020](#)). Synthetic polymeric fibres from fishing gear and nets can result in residue of plastic particles.

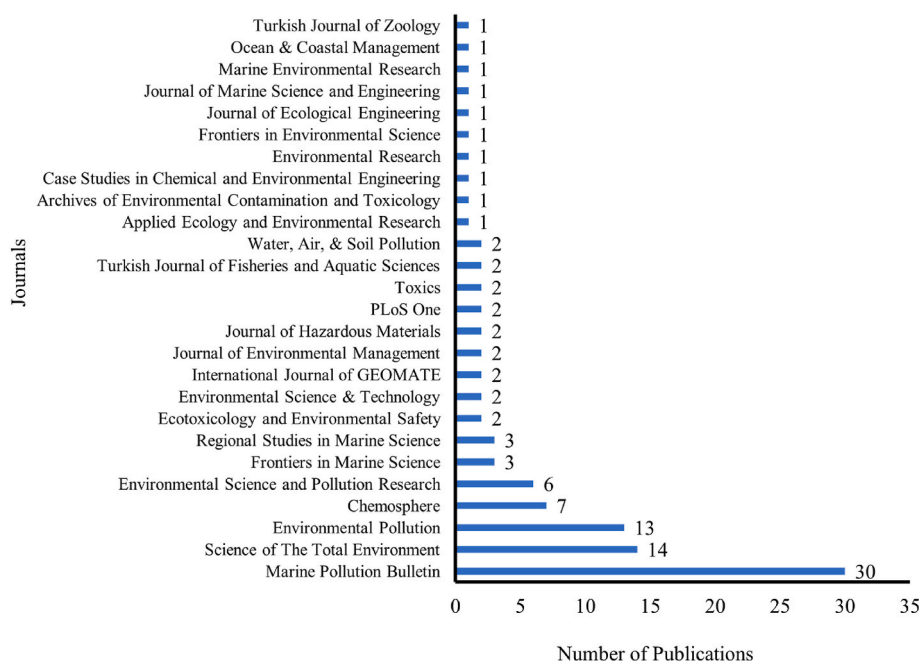


Fig. 1. Number of publications by journals.

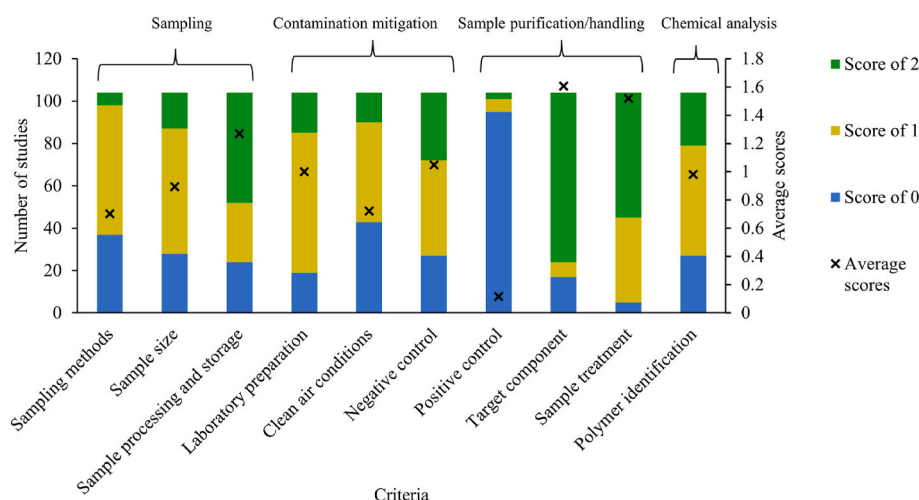


Fig. 2. QA/QC quantitative system scores from 104 studies. The stacked column shows the number of studies in each score value (0, 1 and 2). × represents the average score ($n = 104$) of each criterion.

In addition, the mesh size can affect the size and life stage of fish, which can affect the MP quantity, as larger fish is reported to have a higher MP content compared with smaller fish (Goswami et al., 2020; McIlwraith et al., 2021).

Studies obtained a maximum score of 2 for fully reporting the type of gear, date, location, and depth. For example, da Silva et al. (2022) registered the date, time, position of capture (latitude and longitude) and depth in each fishing event and calculated the distance to coast. Goswami et al. (2020) used a net to trawl for about 30–45 min in the direction opposite to current on 10th August 2018 from six locations. An article with a score of 1 lacks some required information. Studies assigned a score of 1 reported the location and time range/date (Ferante et al., 2022; Wu et al., 2021), or fishing materials, location, time range (Bialowas et al., 2022), or location and fishing materials (Maghsodian et al., 2021). Studies had a score of 0 when they gleaned the samples from local fisherman (Abidli et al., 2021), fish market (Taghizadeh Rahmat Abadi et al., 2021) or simply reported the location

(Sembiring et al., 2020). Most of the studies (59%) scored 1, while 6% of articles scored 2 whereas 36% of studies scored 0. Over half of the studies failed to report a part of the information such as fishing date or materials.

3.2.2. Sample size

Both the International Council for the Exploration of the Sea (ICES, 2015) and the European Strategy Framework Directive's Technical Subgroup on Marine Litter (Hanke et al., 2014) recommend a sample size of at least 50 individuals. Thus, the sample size of 50 is used as a threshold in this criterion. Articles with a sample size of over 50 individuals per species or feeding type score a maximum of 2. For instance, a study in which seventy individuals of each species were collected to investigate the MP ingestion in four species obtained score of 2 (Agharokh et al., 2022). A study with a sample size >50 specimens in total and >25 for each research unit (species, food web, ecoregion, feeding type, etc.) gets a score of 1. The sample size of studies with score

1 ranges from 51 to 1972. For most of the situations, the sample size is sufficient (>50) for the research unit of “fish” but is insufficient (<50 or <25) for each specie or station. For example, Zhang et al. (2021) collected a total of 1003 samples to estimate the MPs ingested by wild fish. The comparison among species or station was carried out, but the sample size in some species or stations were smaller than 50. De Witte et al. (2022) used 71 specimens of 6 species to evaluate the feasibility of biomonitoring MPs in fish gastrointestinal tract (GIT). Although 71 is enough for the research unit of “fish”, each specie is less than 25. A study obtains a score of 0 when its total sample size is less than 50 individuals (Nikki et al., 2021; Sayed et al., 2021). Among the collected articles, 57% scored 1, 27% got 0 and 16% obtained 2. Obviously, the acquired score depends on the research unit considered and its corresponding sample size. Although a study with a higher sample size does not mean a higher score, a larger sample size is recommended as it can lead to a more reliable result. Interestingly, a negative relationship ($r = -0.33$, $P < 0.5$) based on Pearson correlation was found between the sample size and the detection rate (i.e., percentage of fish containing MPs). Intuitively, a larger sample size should provide stronger evidence establishing the extent to which the propensity for a species to ingest MPs.

3.2.3. Sample processing and storage

The highest score is applied to a situation when the fish is stored on ice or frozen immediately after catching it. Any sample processing (e.g., the dissection of fish) is avoided before reaching the laboratory to decrease any possible contamination. Half of the articles obtain a score of 2 in this criterion. For example, Zhang et al. (2021) preserved all samples with aluminium foil at -20°C before analysis and specie identification and biological measurements were performed in the lab to avoid contamination. Goswami et al. (2020) transported samples in an ice-cold box and immediately refrigerated fish at below -20°C upon arriving at the laboratory. 27% of studies got a score of 1 for just providing the information on freeze preservation without providing information on transportation after sampling (Bakir et al., 2020) or dissecting a part of samples in the field, while transporting other samples on ice (Wootton et al., 2021). 23% of articles received a score of 0 since they dissected fish on board (De Witte et al., 2022) or failed to provide any information on sample processing and storage (Sembiring et al., 2020). Further sample handling should be conducted in the lab, especially the dissection, which could raise a higher risk of MP contamination when conducted on board. Only 7 out of 104 studies processed the fish on board, indicating that most of the researchers were aware of the risk of MP contamination in this improper operation.

3.2.4. Laboratory preparation

The precautions in lab preparation include three aspects: washing equipment, tools, and work surfaces; wearing gloves and 100% cotton lab coats; rinsing the exterior of the fish. 18% of the articles had a score of 2 for implementing all these precautions. Naidoo et al. (2020) washed all the fish, glassware and equipment and wore white cotton lab coats. Additionally, they pre-checked Petri dishes and metal forceps under a dissecting microscope for MP contamination. 63% of the studies scored 1 since they missed only one of these aspects. De Witte et al. (2022) did not report rinsing the exterior of the fish. Zhang et al. (2021) did not report washing the tools and equipment. Chen, Fang, et al. (2021) did not report the usage of cotton lab coats. 18% of the reports obtained a score of 0 since they did not provide any information (Sembiring et al., 2020) or just followed one aspect of the precautions (Cabansag et al., 2021). Although these precautions seem easy to follow, only 19 out of 104 studies could meet all the requirements. It can be assumed that most researchers might not be aware of all these three aspects.

3.2.5. Clean air conditions

Airborne MPs are another potential contamination source, and thus it is important to provide a clean air condition for the sample handling and detection procedures. 13% of articles reporting the usage of a

laminar flow cabinet were assigned a score of 2. In terms of using a fume hood or covering the samples and blank controls as much as possible to account for the airborne contamination (Chenet et al., 2021), 45% of studies were labelled with a score of 1 (Siddique et al., 2022). As fume hoods deliver air from outside there is an increased risk of airborne contamination (Wesch et al., 2017). 41% of the reports received the lowest score since they did not provide any information (Maghsodian et al., 2021) or just mentioned a clean environment without giving further details (Taghizadeh Rahmat Abadi et al., 2021). It is ideal for a lab to have a laminar flow cabinet to perform all procedures. If a laminar flow cabinet or a clean room is unavailable, the researchers can follow the precautions such as covering the sample as much as possible, keeping the window closed and running negative samples in parallel.

3.2.6. Negative controls

Even if the researcher obeys the above precautions including sample processing and storage, laboratory preparation and clean air condition, MP contamination might not be absolutely avoided. Therefore, negative controls are essential to verify and correct for contamination. The negative controls should be conducted using filtered water, or biota tissue that is free of plastic and should be performed in parallel to the sample treatment. 31% of the studies scoring 2 in this section performed replicated ($n \geq 3$) procedural blanks. For instance, Hosseinpour et al. (2021) conducted blank samples per working day and for every ten samples to investigate the occurrence of possible airborne MPs. The blank samples showed a mean value of 0.4 MPs/sample. Zhang et al. (2021) ran three control solutions containing 10% KOH throughout the processes to correct potential MP contamination. No MP particles were found in the controls, but natural fibrous particles such as cotton were present. 43% of the studies were assigned to score of 1 as they just examined blank filters (Adika et al., 2020), Petri dishes (Zhang et al., 2021) or ultrapure water (Naidoo et al., 2020) or performed an insufficient number of negative controls (replicates <3) (Ahmadi et al., 2022) or failed to provide the number of the negative controls (Wu et al., 2021). 26% of the articles did not give any information on the negative controls and therefore obtained a score of 0. The evidence that MP contamination was present in the negative controls of some studies (Baalkhuyur et al., 2020; Guilhermino et al., 2021) emphasises the importance for the researchers to perform the blank samples to obtain reliable results. 74% of the articles that have carried out the negative controls indicate that most of the researchers realise the value of this procedure, although some of them conduct less than 3 blank samples.

3.2.7. Positive controls

Positive controls are used to assess MP loss during sample preparation and analysis (Koelmans et al., 2019). Positive controls should be conducted in triplicate with known amounts of MPs added to the samples undergoing the same analytical method. Only three studies performed reliable positive controls (De Witte et al., 2022; McIlwraith et al., 2021; Uurasjarvi et al., 2021). De Witte et al. (2022) performed recovery tests daily by spiking 10 PVC particles (diameter = $251\text{ }\mu\text{m}$) into 20 g samples of mussel tissue. Uurasjarvi et al. (2021) counted fluorescent polystyrene beads (diameter = $90\text{ }\mu\text{m}$) under a fluorescence microscope and transferred the beads into three glass bottles each containing a fish GIT to undergo the enzymatic digestion process. McIlwraith et al. (2021) checked the recovery rate of 5 types of MPs for 3 species of fish. Six studies obtained a score of 1 since they tested less than three replicate positive controls (Reinold et al., 2021; Zhou et al., 2021) or just detected the filter with spiked MPs (Bakir et al., 2020) or just conducted the separation process for the positive samples (Zhang, Cui, et al., 2020) or failed to report the number of positive controls tested (Chen, Fang, et al., 2021; Wu et al., 2020). 91% of the articles scoring 0 for missing information on the positive controls, which is consistent with the finding by Koelmans et al. (2019) and Hermesen et al. (2018), indicating an insufficient application of this criterion for years.

3.2.8. Target component

The choice of the target component depends on the purpose of the study. For accurate assessment of MP digestion, ideally, the entire gastrointestinal tract (esophagus to vent) is required to perform the analysis. For accurately estimating MP contents in other fish parts such as the liver, muscle or brain, it is important to examine the entire organs and tissues. 77% of the studies got the highest score when they used the full specimens (Cordova et al., 2020), or entire GIT (Wootton et al., 2021), or liver, muscle and gill (Guilhermino et al., 2021). The articles using the stomach and intestine as target components were assigned to a score of 1. Analysis of a small part of the organs and tissues received the lowest score. For example, just investigating the stomach (Pradit et al., 2021; Yagi et al., 2022) causes the loss of information on the gut and throat. Selvam et al. (2021) examined 0.5 g of muscle tissue, leading to an underestimation of the MP content in the entire muscle tissue.

3.2.9. Sample treatment

MPs should be extracted from biological samples for subsequent detection and polymer identification. Particularly, sample digestion is required to dissolve organic matter without changing the MP weights, counts or shapes. The 10% KOH or enzyme protocols without heating are proven to minimise the damage to the MPs (Catarino et al., 2017; Dehaut et al., 2016). Based on the study by Munno et al. (2018), the digestion solution is advised to be incubated at or below 60 °C to minimise the loss of any constituent MPs. Additionally, the adequacy of the protocol using 10% KOH solution and incubation at 60 °C was confirmed by Dehaut et al. (2016). Based on these findings, we set 60 °C as the safe upper limit for 10% KOH solution, and as a criterion to assign a maximum score as a precautionary measure, and thus 57% of the studies scored 2 points. 38% of the articles that used other protocols causing damage to the MPs obtained a score of 1. Sembiring et al. (2020), for instance, performed the protocol using hydrogen peroxide to digest samples at 75 °C, which could induce the degradation of MPs (Hurley et al., 2018; Munno et al., 2018). Khan et al. (2020) applied the NaOH protocol that resulted in adverse destructive effects on the integrity of MPs (Dehaut et al., 2016; Hurley et al., 2018). Four studies got a score of 0 for just separating MPs from samples without digestion when some particles were smaller than 300 µm.

3.2.10. Polymer identification

Polymer identification by using analytical techniques such as Fourier transform infrared spectroscopy (FTIR), Raman, pyrolysis gas chromatography-mass spectrometry (GC-MS) and scanning electron microscope/energy dispersive X-ray spectra (SEM/EDX) (Gniadek & Dąbrowska, 2019; Hermesen et al., 2018) enables identification of polymers and distinguishes them from non-polymeric materials, providing a more reliable assessment of MP prevalence. To investigate the distribution of polymer types among samples, it is advised to obtain a representative subsample to undergo the polymer identification (Hermesen et al., 2018; Koelmans et al., 2019). A representative subsample is regarded as all particles when the number of microscopically pre-sorted particles is < 100. For particle numbers >100, >50% of pre-sorted particles should be used as a subsample with a minimum of 100 particles. 24% of studies got 2 points for performing the identification with a representative subsample as described above. 50% of articles identified the particles in an insufficient way or did not provide the numbers and thus got a score of 1. 26% of studies obtained a score of 0 when they only performed the visual sorting without polymer identification.

3.2.11. Overall reliability of method aspects and studies

Total accumulated scores (TAS), as shown in Excel file format in the Supplementary Material, ranged from 4 to 16 out of a maximum of 20. The average value of TAS for 104 studies were roughly half (9.8/20) of the maximum TAS, which is very similar to the average score (8.41/18) of studies in freshwaters and drinking water reported by Koelmans et al.

(2019). In the current review, we found TAS in 50% of studies ranging from 4 to 10 (≤ 10), in 41% of articles between 10 and 15, and in 2 studies ≥ 15 . Only two studies received positive score values for all criteria (Bakir et al., 2020; McIlwraith et al., 2021). Studies obtained a low score due to missing information on certain criteria, most notably positive controls. Average scores per defined criterion were lower than 1 for the positive controls (0.12), sampling method (0.70), clean air conditions (0.72), sample size (0.89), and polymer identification (0.98) (Fig. 2). Due to the lack of information on the quality aspects discussed above, a study can become irreproducible and less reliable, and thus a significant improvement is required especially for these five criteria.

It was suggested by de Ruijter et al. (2020) that a study with higher scores provides more reliable and useful data in the context of risk assessment. TAS was graded by Gouin (2020) to reflect a more general trend, with a TAS ≤ 10 representing that a study failed to meet the requirement of the assessment criteria, a TAS between 10 and 15 implying that study results were of moderate quality, and a score ≥ 15 indicating that the study were of high quality to satisfactorily addressed all the criterion. However, we must stress here that TAS, while being a useful method to make comparisons among studies based on common characteristics, it is not an absolute judgment to be used as a ranking list of the value of studies (Hermesen et al., 2018). It is essential to critically evaluate each study upon its own merits, plausibility, and comparability to other studies.

3.3. Characteristics of microplastics in fish

3.3.1. Global distribution of microplastic contamination in fish

The reviewed 104 articles investigated MP ingestion by fish in 44 countries and regions, with most of the research papers distributed in five countries, i.e., China (23%), Iran (8%), Thailand (5%), Turkey (4%), Indonesia (4%), while half of them pertained to 39 other countries and regions (Fig. 3A). Similarly, Sequeira et al. (2020) indicated that their selected works (from March 2019 to March 2020) pertained to 24 countries, with most studies coming from China (29%). Among forty-seven articles reported the percentage of fish containing MPs, twenty-two studies with TAS >10 were selected to generate the global map showing the distribution of microplastic contamination in fish (Fig. 3B). The percentage of fish containing MPs varied from 6% to 94%, with a median of 41%. No significant differences were found among continents, with the order of median value: Africa (52%) > Europe (44%) > Asia (40%) > Oceania (36%). Sequeira et al. (2020) also reported no significant difference among continents based on 33 studies. Therefore, it is better to include more data to evaluate the difference among continents.

3.3.2. Polymer types

Out of the 104 articles studied, 76 reported the polymer types of detected MPs. The five most frequently observed polymer types across the investigated studies are Polyethylene (PE) > Polyamide (PA, including nylon) > Polypropylene (PP) = Polyethylene Terephthalate (PET) > Polystyrene (PS) (Fig. 4A). Previous reviews (Lim et al., 2022; Sequeira et al., 2020; Zantis et al., 2021) regarding the ingestion of MPs by fish also demonstrated the major polymer types including PE, PA, PP, PS, PET, and PA, with PE being the predominant one, which is consistent with the current review. The largest groups in total non-fibre plastics production are PE (36%), PP (21%), and Polyvinyl Chloride (12%), followed by PET, Polyurethane, and PS, with PE contributing most to pollution in all environments (Geyer et al., 2017). PE, PP and PS are widely used for fishing tools and packaging materials of foods and manufactured products, such as plastic films, plastic bags, bottle containers, etc. (Kalaiselvan et al., 2022; Lim et al., 2022; Makhdomi et al., 2021). PET has diverse applications in packaging, textile, and films (Deshmukh & Bhat, 2003). PA is commonly applied in fishery activities and the clothing industry, as well as in the process of many nylon product manufacturing (Lim et al., 2022; Makhdomi et al., 2021).

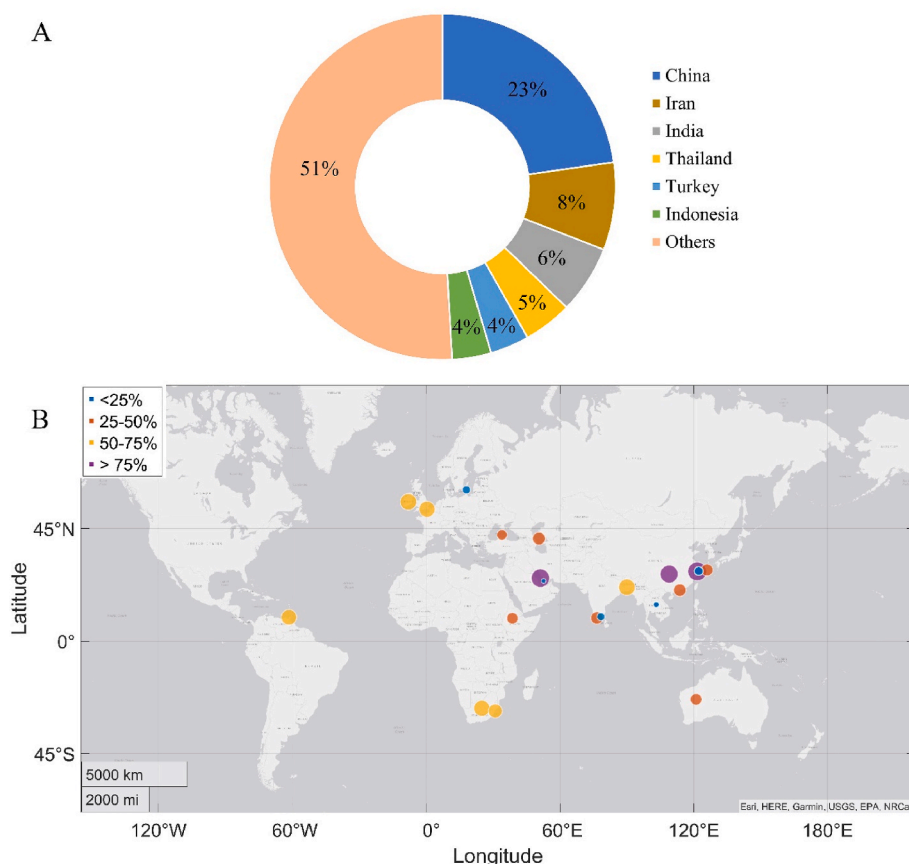


Fig. 3. (A) The percentage of articles performed by countries, (B) the global map showing the distribution of MP contamination in fish. Circles in different colours indicate different percentages of fish contaminated by MPs.

These plastic materials can accumulate in the marine environment as primary and secondary plastics through accidental loss, sewage inflow, fishing, and recreational activities (Kalaiselvan et al., 2022).

3.3.3. Size

Generally, MPs were divided into two groups called small MPs (<1 mm) and large MPs (1–5 mm) (Naji et al., 2019). Among 62 studies reporting the predominant size range of MPs, 57% of the studies recorded small microplastic as the major size class (Fig. 4B). The predominance of small MPs might be due to the following reasons. Firstly, small MPs are close to the size of food (Ory et al., 2017) and are relatively easy to mix into food (Koongolla et al., 2020), resulting in preferential capture and ingestion by fish. Secondly, fish may decrease the MP size through the mechanical action of chewing and ingesting, which can break large MPs down into smaller ones (Zhang et al., 2021). Furthermore, small MPs tend to retain longer in fish GIT, and are more easily translocated into tissues and organs compared to larger MPs (Roch et al., 2021). In conclusion, small particles are more likely to be ingested accidentally by fish and tend to be transferred to different organs and tissues, inducing a potential threat to fish and human health (Koongolla et al., 2020).

3.3.4. Colour

Among 72 studies provided the information on the major colour of MPs in fish, blue is the most common colour, following by black, transparent, white, red, and green (Fig. 4C). Similarly, two previous reviews reported that blue and black are two predominant colours of MPs ingested by fish (Lim et al., 2022; Zantis et al., 2021). For the reports detecting MPs in surface water, the primary colours are black (Özhan Turhan, 2021); green and white (Sayed et al., 2021); white, yellow and black (Zhang, Cui, et al., 2020); black, red and blue

(Agharokh et al., 2022). For studies conducting the detection of MPs in sediment, the predominant colours were transparent and black (Özhan Turhan, 2021); green and white (Sayed et al., 2021); white (Maghsodian et al., 2021); transparent and blue (Merga et al., 2020); black, red and blue (Agharokh et al., 2022). The commonly reported predominant colours in surface water and sediment are black, transparent, white, and blue. Accidental ingestion by breathing in surface water contaminated by MPs or feeding on the food attached to MPs or predation of their prey ingesting MPs are the possible major sources of the MPs in fish (Lim et al., 2022; Merga et al., 2020). Therefore, the abundance of colour ingested by fish might be due to the similar predominant colour in sediment and water surface. However, Agharokh et al. (2022) found no significant correlation between the colour of MPs in surface water-fish and fish-sediment. Another explanation of the MP source is that the specific fish species intentionally ingested colourful plastic resembling their prey (Lim et al., 2022; Merga et al., 2020), which is supported by a phenomenon that blue pigmented organism was predominant in ocean zooplankton (Herring, 1965). Therefore, the major colour of MPs can vary among different fish species in the same habitat (Agharokh et al., 2022) due to the different ingested strategies of the MPs.

3.3.5. Shape

The pooled data indicated that the typical shapes of MPs ingested by fish were fibre (77%), fragment (18%), film (2%), pellet (1%), sphere (1%), and granule (1%) (Fig. 4D). The shapes of the detected MPs in fish are mainly affected by the ingestion and egestion of fish (Lim et al., 2022). Fish ingest plastics randomly through feeding and breathing and thus the distribution of shapes is similar to the surface water and sediment (Rocha-Santos & Duarte, 2017). Several studies reveal that the main shapes of MPs in surface water and sediment are fibre, fragment and film (Agharokh et al., 2022; Maghsodian et al., 2021; Merga et al.,

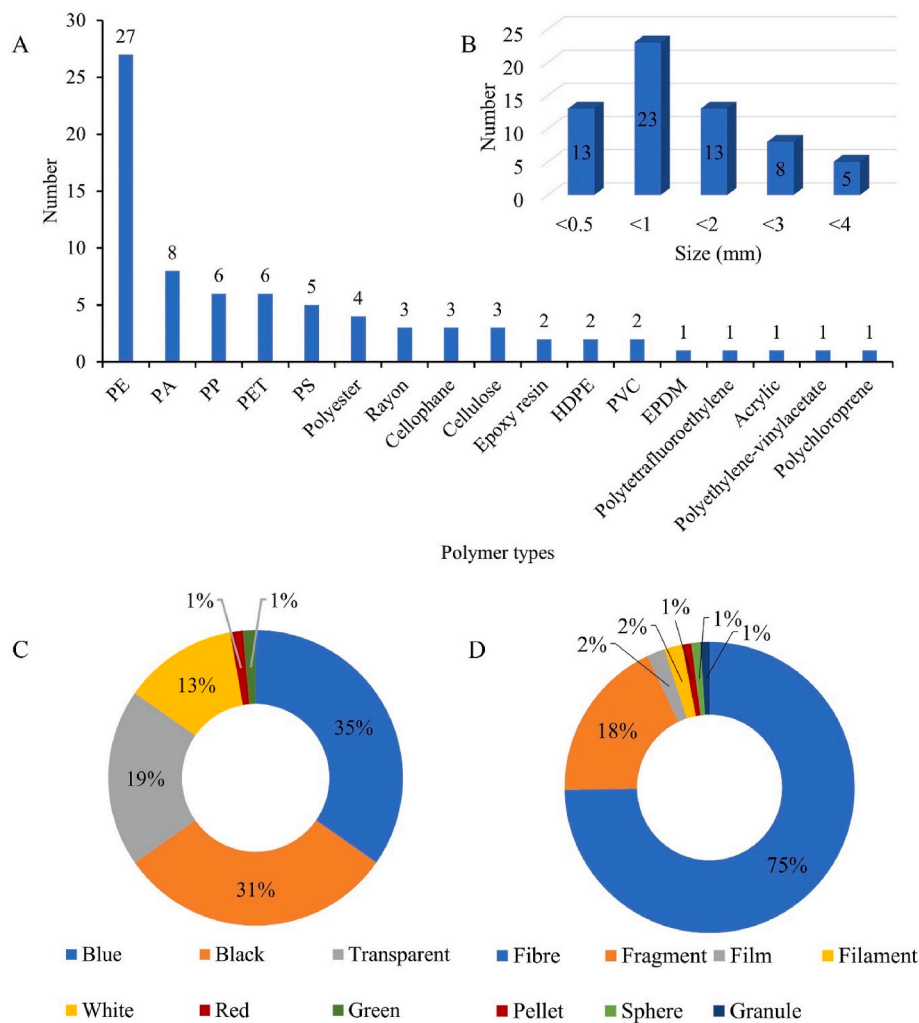


Fig. 4. Characteristics of microplastic in fish: (A) Number of studies reporting the major polymer type (76 out of 104 records reported major polymer type), (B) Number of studies reporting the major size (62 out of 104 records reported major size), (C) Percentage of the major colour (72 out of 104 records reported major colour), (D) Percentage of the major shape (99 out of 104 records reported major shape). PE = Polyethylene; PP = Polypropylene; PA = Polyamide; PS = Polystyrene; PET = Polyethylene Terephthalate; PVC = Polyvinyl Chloride; HDPE = High density polyethylene; Nylon = polyamide 6; EPDM = Ethylene Propylene Diene Monome.

2020; Sayed et al., 2021; Özhan Turhan, 2021), which are in accordance with the dominant shapes in fish. Furthermore, fragments are more difficult to adhere to organic items compared to fibre plastic particles, which might reduce the accident ingestion by fish (Merga et al., 2020). On the other hand, Xiong et al. (2019) hypothesised that most MPs could be egested through excretory processes rather than accumulated in the gut. Some shapes of MPs may prevent their egestion and are more likely to be detected (Rocha-Santos & Duarte, 2017). For example, the egestion rate of filament in goldfish was lower than film and fragment (Xiong et al., 2019).

4. General discussion

Quality assessment was used to evaluate the reliability of 104 studies regarding MP ingestion by fish. Some common problems still exist since Hermesen et al. (2018) proposed the quality criteria for analysing MPs in biota samples. Almost all the studies (98%) failed to be fully reliable as they scored 0 in at least one criterion, implying uncertainty appeared in some aspects. Particularly, 91% of the studies scored 0 for not performing the positive controls. Compared with other criteria, the positive control is appreciated less or ignored by researchers, which might be due to the assumed reliability of the digestion method or due to the lack of the standard MPs. However, positive controls are essential to improve the accuracy of an experiment since uncertainty can appear in different circumstances. Another underestimated criterion by researchers is clean air conditions. Even if some labs might suffer from the lack of the laminar flow cabinet, they can minimise contamination by covering the

samples and performing the negative controls. However, 41% of the studies reviewed here obtained a score of 0 for not taking any action to provide a clean air condition or not providing the relevant information. The proposed quality criteria for the analysis of MP in biota samples might not be known by some researchers, since about 15% of the studies got TAS less than 6 for failing to meet most of the criteria. Nevertheless, there are two criteria that might be difficult to achieve, namely, sampling methods and polymer identification. For some researchers, they obtain the fish from a third party without instruction on handling of organisms, which is not inherently wrong, provided that the interest of the study lies in contamination of seafood, not in natural populations. In addition, some researchers might not be able to perform MP identification due to the lack of specific equipment such as FTIR, Raman, pyrolysis GC-MS and SEM/EDX.

Most of the studies evaluated herein used GIT as the target component to investigate MPs digested by fish. Nevertheless, certain fish species are consumed as a whole and the MPs in their GIT can be directly exposed to a human. Generally, the muscle, liver and fillet are served as food, resulting in the direct consumption of MPs by humans. For the current review, fourteen out of the 104 studies evaluate the MP content of muscles. MPs transferred to the muscle or other organs are smaller than those in the GIT, making it more difficult to detect compared to the MPs in GIT. As reported by Koongolla et al. (2020), MP size in the muscle of six fish species ranged from 1.6 to 2.8 μm . The most common methods to identify MPs are Raman and FTIR which are unsuitable for the detection of plastic particles below 20 μm . Koongolla et al. (2020) applied SEM-EDX technology to detect the MPs less than 3 μm , while

Zhang et al. (2021) adopted Raman tweezers to investigate MPs less than 20 µm. Therefore, the restrictions of detection methods would be an obstacle for some researchers to evaluate the MPs in muscles or other organs.

The uptake of MPs by fish is unavoidable due to confusion with food, bioaccumulation from the food chain and accidental ingestion while drinking water and breathing (Roch et al., 2020). Several factors, including the plastics in the habitat, the intake and egestion of MPs by fish can affect the size, colour and shape detected in fish. Major colours and shapes in fish were found to be consistent with their habitat (surface water and sediment) (Agharokh et al., 2022; Jaafar et al., 2021; Merga et al., 2020), which might be attributed to the random intake and accident ingestion of MPs by fish. However, there was no significant correlation between shape, size, and colour of MPs in surface water-fish, fish-sediment, and sediment-surface water (Agharokh et al., 2022). Therefore, more studies are encouraged to investigate the correlation between the MPs in fish and their habitat, since other factors could also affect MP type in fish. Firstly, MPs in small size and fibre shape are more likely to mix with food, inducing more unintentional ingestion of these MPs. Secondly, some fish species are visual predators, preying on small zooplankton and may be confused by MPs resembling their prey in shape, colour and size (Mizraji et al., 2017; Ory et al., 2017; Walkinshaw et al., 2020). Thirdly, egestion rate and retention time of MPs varied in different shapes and size. The excretion was affected by the MP size, which are proved by the evidence that a faster excretion rate for 125–150 µm PE microbeads than for 65–75 µm PE microbeads (Hoang & Felix-Kim, 2020). Microplastic fibres retained in GIT significantly longer than other shapes (Au et al., 2015; Xiong et al., 2019). Moreover, microplastic fibres could be found in the muscle tissue of fish (Akhbarizadeh et al., 2019), while translocation of spherical microplastics (10–300 µm) was not observed after dietary administration to adult rainbow trout (Kim et al., 2020). The patterns of MP excretion differ in various fish species, however, reports on MP retention and excretion times in fish are scarce (Okamoto et al., 2022). Therefore, studies about egestion and translocation of MP in varieties of fish species need further investigation to better understand the MP threat to fish and human health.

5. Conclusions

Around half of the studies in the current review failed to meet the requirement of the quality assessment criteria proposed by Hermesen et al. (2018) while the rest were of moderate quality, implying that the reliability of these studies is not at such a level that could be used confidently for monitoring microplastics in fish. Although the quality criteria system can be applied to control the quality of a study and enable easier comparison of studies to move toward standardisation and reliability, some items such as sampling method, positive controls and polymer identification are difficult for some researchers to achieve. Hence, when using the criteria systems, researchers are encouraged to optimise the difficult criteria and perform the remaining criteria as best as they can to assure the reliability and reproducibility of their findings. Despite the quality limitations, the pooled data confirmed that microplastics are frequently present in various fish species. The major size range, polymer type, colour, and shapes of microplastics in fish were 0–1 mm, polyethylene, blue, and fibre, respectively, which depended on the microplastic distribution in fish habitat, the uptake and excretion of microplastics by fish. Further studies are required to investigate the exceedingly small microplastics in fish muscles, livers, and other eatable tissues to better understand the translocation of microplastic in fish and their uptake possibility by humans.

CRediT authorship contribution statement

Xiaohui Lin: Methodology, Validation, Formal analysis, Investigation, Writing – original draft. Aoife A. Gowen: Validation, Resources,

Writing – review & editing, Visualization. Hongbin Pu: Formal analysis, Writing – review & editing. Jun-Li Xu: Validation, Resources, Supervision, Writing – review & editing, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodcont.2023.109939>.

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